

LinkCutter — Test Report

ICCS423 Computer Security — A02: Mini Open Source
Cybersecurity Tool

Test Information

Test Date	2026-07-02
Tester	Aung Htin Kyaw(Goon)
Tool Version	0.1.0
Commit Hash	61c21f079706c58822cfffbb39e81cf34144c1b6
Rust Version	rustc 1.96.0 (ac68faa20 2026-05-25)
OS	macOS (Apple Silicon M2)

Automated Test Results

LinkCutter includes 44 automated tests: 33 unit tests across the scanner and shortener modules, and 11 integration tests verifying end-to-end behavior. All tests were run using cargo test.

```
* cargo test
  Finished `test` profile [unoptimized + debuginfo] target(s) in 0.04s
  Running unittests src/lib.rs (target/debug/deps/url_threat_shortener-affd05739ba33057)

running 33 tests
test scanner::tests::test_at_symbol_normal_url ... ok
test scanner::tests::test_at_symbol_trick_detected ... ok
test scanner::tests::test_excessive_length_normal_url ... ok
test scanner::tests::test_ip_literal_normal_domain ... ok
test scanner::tests::test_ip_literal_ipv4 ... ok
test scanner::tests::test_excessive_length_many_subdomains ... ok
test scanner::tests::test_known_bad_empty_blocklist ... ok
test scanner::tests::test_excessive_length_long_url ... ok
test scanner::tests::test_known_bad_no_match ... ok
test scanner::tests::test_known_bad_match ... ok
test scanner::tests::test_normalize_preserves_existing_scheme ... ok
test scanner::tests::test_normalize_trims_whitespace ... ok
test scanner::tests::test_normalize_adds_scheme ... ok
test scanner::tests::test_punycode_detected ... ok
test scanner::tests::test_punycode_in_subdomain ... ok
test scanner::tests::test_scan_url_clean ... ok
test scanner::tests::test_punycode_normal_domain ... ok
test scanner::tests::test_scan_url_empty_input ... ok
test scanner::tests::test_scan_url_invalid_input ... ok
test scanner::tests::test_scan_url_without_scheme ... ok
test scanner::tests::test_suspicious_tld_flagged ... ok
test scanner::tests::test_suspicious_tld_safe ... ok
test scanner::tests::test_typosquat_exact_match_not_flagged ... ok
test scanner::tests::test_scan_url_multiple_flags ... ok
test shortener::tests::test_generate_code_length ... ok
test shortener::tests::test_generate_code_uniqueness ... ok
test shortener::tests::test_load_nonexistent_file ... ok
test shortener::tests::test_resolve_not_found ... ok
test scanner::tests::test_typosquat_unrelated_domain ... ok
test scanner::tests::test_typosquat_detected ... ok
test scanner::tests::test_scan_url_high_risk ... ok
test shortener::tests::test_save_and_load ... ok
test shortener::tests::test_resolve_found ... ok

test result: ok. 33 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

  Running unittests src/main.rs (target/debug/deps/linkcutter-c41db705c9ae372e)

running 0 tests

test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

  Running tests/integration_test.rs (target/debug/deps/integration_test-cbe1c2041d9f3413)

running 11 tests
test test_full_scan_invalid_input ... ok
test test_load_blocklist_missing_file ... ok
test test_full_scan_clean_url ... ok
test test_full_scan_at_trick ... ok
test test_full_scan_empty_input ... ok
test test_full_scan_auto_normalize ... ok
test test_full_scan_ip_literal ... ok
test test_full_scan_multiple_flags ... ok
test test_resolve_nonexistent_code ... ok
test test_shorten_resolve_roundtrip ... ok
test test_multiple_mappings ... ok

test result: ok. 11 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s

  Doc-tests url_threat_shortener

running 0 tests

test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00s
```

Manual Test Cases

T1 — Scan a clean URL

```
> linkcutter scan "https://google.com"
```

URL Threat Scanner & Shortener

[LOW RISK] Score: 0/7

No threats detected.

T2 — Scan a URL with suspicious TLD

```
> linkcutter scan "https://free-prize.tk/claim"
```

URL Threat Scanner & Shortener

[MEDIUM RISK] Score: 1/7

Flags:

```
! Suspicious TLD detected: .tk
```

T3 — Scan a typosquat domain

```
> linkcutter scan "https://paypa1.com/login"
```

URL Threat Scanner & Shortener

[MEDIUM RISK] Score: 1/7

Flags:

```
! Possible typosquat: "paypa1.com" is 1 edit(s) from "paypal.com"
```

T4 — Scan an IP literal URL

```
> linkcutter scan "http://192.168.1.1/admin"
```

URL Threat Scanner & Shortener

[MEDIUM RISK] Score: 2/7

Flags:

```
! URL uses raw IPv4 address: 192.168.1.1
```

```
! Domain has 4 levels (e.g. "192.168.1.1" - threshold: 3)
```

T5 — Scan a URL with @ symbol trick

```
> linkcutter scan "https://google.com@evil.com/steal"
```

```

      _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
      / /  ( ) _ _ _ / / _ / _ _ _ / _ _ / / / / _ _ _ _ _ _ _ _ _ _ _ _
      / /  / / _ V // _ / /  / / / / _ / _ V _ _ _ /
      / / _ _ / / / / / , < / / _ _ / / _ / / / / _ / _ _ /
      / _ _ _ _ / / _ / / _ / \ _ \ _ _ _ \ _ _ , \ _ \ _ \ _ \ _ _ _ / /

```

URL Threat Scanner & Shortener

[MEDIUM RISK] Score: 1/7

Flags:

```
! URL contains '@' symbol - the real destination may be hidden (userinfo trick)
```

T6 — Scan invalid input

```
> linkcutter scan "not a url"
```

```

      _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
      / /  ( ) _ _ _ / / _ / _ _ _ / _ / / / / _ _ _ _ _ _ _ _ _ _ _ _
      / /   / / _ \ // _ / /   / / / / _ / _ / _ \ _ _ /
      / / _ _ / / / / / , / / _ _ / / / / / / / / _ / /
      / _ _ _ _ / _ / / _ / \ _ \ _ _ _ \ _ _ , _ \ _ \ _ \ _ _ _ / /

```

URL Threat Scanner & Shortener

```
Error: Invalid URL: "not a url" is not a valid URL. Example: https://example.com
```

T7 — Scan empty input

```
> linkcutter scan ""
```

```

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
//  ( )__ //_/ ___/_ _/ // /___/ ___/ ___/
//  // _ V //_/ // / / / _/ _/ _ V ___/
// ___/ / / / / ,< / ___/ / / / / / _/ /
/___/ / / / / / \ \ \___/ \ _ , \ _ \ _ \ ___/ /

```

URL Threat Scanner & Shortener

```
Error: Invalid URL: URL cannot be empty
```

T8 — URL auto-normalization (missing scheme)

```
> linkcutter scan "google.com"
```

```

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
//  ( )__ ///_/ ___/_ _// // /_/_ _ _ _ _ _ _ _ _ _ _ _ _ _ _
//  // _ \ // // // // // // // // // // // // // // // // // //
//  _// // // // // // // // // // // // // // // // // // // //
/_ _ _ _ _/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/_/

```

URL Threat Scanner & Shortener

[LOW RISK] Score: 0/7

No threats detected.

T9 — Shorten and resolve round trip

```
> linkcutter shorten "https://hello.com"

  _  _  _  _  _  _
 / /  ( )__ / /_ / ___/_ _ / /_ / ___ _
 / /  / / _ V //_ / / / /_ / _ / _ V ___/
 / /___ / / / / / ,< / /___ / /_ / /_ /_ /
 /_____/ / / / /_ / \___ / \_ , / \_ / \_ / \_ /

URL Threat Scanner & Shortener

[LOW RISK] Score: 0/7
No threats detected.

Short code: 4fIV5d
Stored successfully.
> linkcutter resolve 4fIV5d

  _  _  _  _  _  _
 / /  ( )__ / /_ / ___/_ _ / /_ / ___ _
 / /  / / _ V //_ / / / /_ / _ / _ V ___/
 / /___ / / / / / ,< / /___ / /_ / /_ /_ /
 /_____/ / / / /_ / \___ / \_ , / \_ / \_ / \_ /

URL Threat Scanner & Shortener

Code:      4fIV5d
URL:       https://hello.com
Risk:      Low (0/7)
Created:   2026-07-17T04:15:58Z
```

T10 — Resolve nonexistent code

```
> linkcutter resolve doesnotexist
```

URL Threat Scanner & Shortener

Error: Short code "doesnotexist" not found

T11 — Interactive mode

```
> linkcutter
```

URL Threat Scanner & Shortener

```
Welcome to LinkCutter! Run with --help for direct command usage.
```

What would you like to do?:

> Scan a URL

Shorten a URL

Resolve a short code

Exit

T12 — Multiple flags (high risk)

[illegible]

Summary Table

Test	Description	Status
T1	Clean URL scan	Pass
T2	Suspicious TLD detection	Pass
T3	Typosquat detection	Pass
T4	IP literal detection	Pass
T5	@ symbol trick detection	Pass
T6	Invalid input handling	Pass
T7	Empty input handling	Pass
T8	Auto-normalization (missing scheme)	Pass
T9	Shorten + resolve round trip	Pass
T10	Resolve nonexistent code	Pass
T11	Interactive mode	Pass
T12	Multiple flags / high risk	Pass

Known Bugs and Limitations

The following limitations are known and documented:

1. Typosquat detection is limited to a built-in list of approximately 15 popular domains. A domain like "faceb00k.com" may not be flagged if it falls outside the Levenshtein distance threshold of 2 from the reference list. A production version would use a much larger reference set.
2. The timestamp in stored URL mappings is approximate. The tool calculates timestamps from UNIX epoch without timezone handling or a date library. A production version would use the chrono crate for accurate timestamps.
3. JSON storage is not optimized for scale. The store file is fully read and rewritten on every save. This is suitable for small-scale use but would not work for high-throughput production scenarios.
4. No live URL fetching. The tool analyzes the URL string itself, not the content at the destination. A URL that passes all heuristic checks could still host malicious content.
5. ANSI color codes in the banner may not render correctly on terminals that do not support 256-color mode. This is cosmetic only and does not affect functionality.